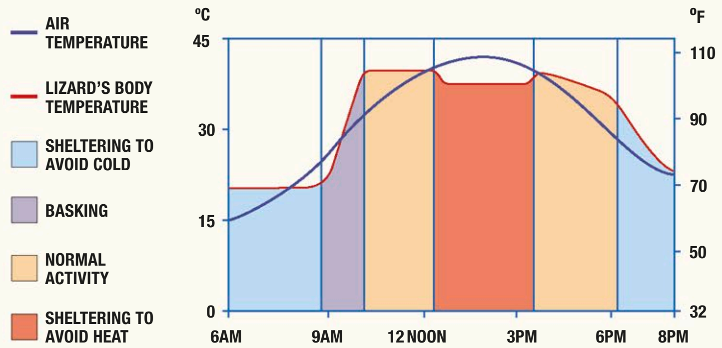
ACTIVITY PATTERNS

Maintaining an optimum body temperature is the key to a reptile's survival and lifestyle, as this graph depicting a diurnal lizard's activity patterns shows. At night and in the early morning, the animal shelters from the cold in its burrow. Later, as the temperature rises, it has to bask in the sun in order to obtain the energy required to forage. The lizard must seek shelter around noon to prevent overheating, but re-emerges later, as the air cools, to forage once more.



VESTIGIAL LIMBS
In some snake families, such as dwarf boas and some boas and pythons, the vestigial back limbs are still evident, indicating an evolutionary link between snakes and lizards.

end when sexual maturity is reached, which means that some long-lived adults can grow extremely large.

Senses

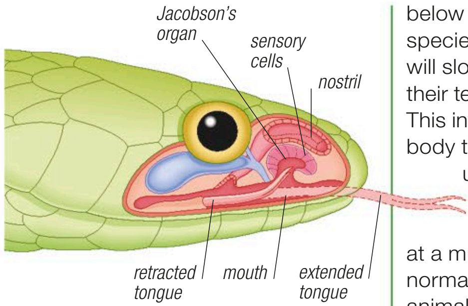
Reptiles' senses are better developed than those of amphibians, and some have sense organs that are not found anywhere else in the animal kingdom. The eyes are often large and well developed, although many snakes have poor sight, and in some burrowing squamates the eyes are reduced or absent. Turtles and tortoises, crocodilians, and most lizards have mobile eyelids, while in snakes and some lizards they are immovable. Lizards and tuataras have a light-sensitive area on top of their skull, known as the third eye, which is thought to control diurnal and seasonal patterns of activity by measuring day length. Reptiles tend

EYE COVERINGS

Crocodilians, turtles, tortoises, and most lizards have 2 movable eyelids (an upper and a lower) as well as a nictitating membrane. This membrane consists of a transparent fold of skin that is drawn over the eye from the side, providing protection while also allowing the animal to see.

In snakes and some lizards, the lower lid, which is transparent, is fused with the upper one, forming a fixed, transparent protective covering over the eye known as the spectacle, or brille.

SHEDDING SPECTACLES
A snake (here, a cross-marked sand snake) sheds the outer layer of its spectacles, or brilles, at the same time as it sloughs the rest of its skin.



JACOBSON'S ORGAN
Snakes and most lizards extend and flicker their tongue to pick up scent molecules around them. On retracting their tongue, they transfer the molecules to the Jacobson's organ, inside the mouth, where the scents are analysed.

to have poor hearing. Some have no external ear opening or middle ear structure at all, and transmit sounds through the skull bones. Taste is not important to reptiles, but smell is highly developed. Some snakes have heat-detecting organs in their faces, and can detect minute temperature changes to help them to locate prey.

Temperature control

Since reptiles cannot generate heat internally, they depend on external factors to keep their temperature within critical limits. At temperatures below their preferred range (in most species, 86–104°F/30–40°C), they will slow down and may act to raise their temperature, perhaps by basking. This involves flattening or angling their body toward the sun, or pressing their underside against a warm rock.

At very low temperatures, their bodily functions operate at a much reduced rate, although normally this occurs only after the animal has sought shelter, in a burrow or under rocks. In temperate regions, reptiles may shelter for prolonged periods, or hibernate, over winter. Similarly, species from hot, arid climates may shelter for the hottest part of the year, an activity known as estivation. Reptiles that live in tropical regions rarely need to bask.

Reproduction

Little is known about the courtship behavior of reptiles, although chemical communication probably plays a large role. Several reptiles vocalize during the breeding season, and males of many lizards and some other groups indulge in visual displays using bright colors, crests, and flaps of skin (dewlaps) under the chin. These displays serve to establish territories as well as to attract females. In most cases, a female is fertilized by a male, although parthenogenesis, in which a female reproduces without the need for fertilization, occurs in several species of lizards and one species of snake.

Most reptiles lay eggs on land, although a significant number of lizards and snakes give birth to live young. Reptile eggs may have a hard shell, like those of birds, but most have a leathery shell that allows water and oxygen to pass through to the developing embryo. Reptiles hide their eggs in burrows, decaying vegetation, or other similar locations. Incubation periods can last from a few days to several months, with the young of some species overwintering in their nest and emerging nearly one year later. Live-bearing species retain the eggs inside their body and in some cases nourish them through a form of placenta. Hatchling and newborn reptiles are very similar to their parents, although their colors and markings may differ. Parental care is rare except in crocodilians, where it may last for 2 years or more. Some lizards also care for their young after birth.

EMERGING FROM HIBERNATION
A group of red-sided garter snakes emerge from their communal burrow, where they have been in hibernation since the onset of cold weather in winter.

GIVING BIRTH
The female viviparous, or common, lizard retains her eggs inside her body until they are ready to hatch. The 2–12 young break out of their eggs as soon as they are outside their mother's body and can survive without her care.